

Email Phishing Analysis Report

Cybersecurity Internship – Task 3

Project Title

Email Risk Analyzer

Prepared By

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Internship: Cybersecurity Internship

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1. Objective

The objective of this project is to analyze an email for phishing indicators by examining both the email headers and message body. The analyzer identifies suspicious authentication results, phishing-related content, spoofed domains, and malicious links before calculating an overall phishing risk score.

2. Email Information

Field	Value
Sender	security@paypa1-support.com
Recipient	user@example.com
Subject	Urgent: Your Account Has Been Suspended
Email Type	Sample Phishing Email

3. Header Analysis

Header	Result
Received	Present
Return-Path	Suspicious
SPF Status	Failed
DKIM Status	Missing

Findings

- SPF authentication failed.
- DKIM signature is missing.
- Return-Path does not appear trustworthy.

- Multiple indicators suggest sender spoofing.

4. Body Analysis

The email body was scanned for phishing indicators.

Suspicious Elements Found

- Urgent action requests
- Account suspension warning
- Threat of permanent account closure
- Suspicious hyperlink
- Fake PayPal domain
- Social engineering language

Suspicious URL

<http://paypa1-support.com/verify-account>

The domain uses "1" instead of the letter "l", indicating a spoofed website.

5. Phishing Indicators

Indicator	Status
SPF Failure	Detected
Missing DKIM	Detected
Suspicious Return-Path	Detected
Suspicious URL	Detected
Urgent Language	Detected
Domain Spoofing	Detected

6. Risk Score

Metric	Value
Risk Score	82 / 100
Risk Level	HIGH RISK

7. Security Assessment

The analyzed email exhibits several common phishing characteristics.

These include:

- Failed SPF authentication
- Missing DKIM signature
- Suspicious sender information
- Spoofed domain name
- Fake verification link
- Urgent language encouraging immediate action

Based on these findings, the email has been classified as a **High Risk Phishing Email**.

8. Recommendations

- Do not click any links contained in the email.
 - Do not download any attachments.
 - Verify the sender through an official communication channel.
 - Check SPF, DKIM, and DMARC authentication before trusting the email.
 - Report the email to the organization's security team.
 - Delete the email if it is confirmed to be malicious.
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9. Conclusion

The Email Risk Analyzer successfully detected multiple phishing indicators by analyzing the email headers and body content. The tool assigned a high phishing risk score and generated a security assessment, demonstrating how automated email analysis can help identify phishing attempts and improve cybersecurity awareness.

10. Screenshots

Insert the following screenshots:

- **Figure 1:** Program Execution Output
 - **Figure 2:** Email Header Analysis
 - **Figure 3:** Risk Score and Classification
 - **Figure 4:** Final Security Report
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Project: Email Risk Analyzer

Language: Python 3

Purpose: Educational Cybersecurity Internship Project